



MATH110: Regression Correlation

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Learning Objectives

- Interpret the correlation coefficient r
- Compute the slope and intercept of the least-squares line
- Write and use a regression equation for prediction

Simplify each expression completely. Show all steps and circle your final answer.

Correlation Coefficient Interpretation

1. A study yields a correlation coefficient of $r = 0.78$. Describe the direction and strength of the linear relationship.

$$r = 0.78$$

Answer: _____

2. A study yields a correlation coefficient of $r = 0.54$. Describe the direction and strength of the linear relationship.

$$r = 0.54$$

Answer: _____

3. A study yields a correlation coefficient of $r = -0.65$. Describe the direction and strength of the linear relationship.

$$r = -0.65$$

Answer: _____

4. A study yields a correlation coefficient of $r = -0.28$. Describe the direction and strength of the linear relationship.

$$r = -0.28$$

Answer: _____

5. A study yields a correlation coefficient of $r = -0.92$. Describe the direction and strength of the linear relationship.

$$r = -0.92$$

Answer: _____

Least-squares regression line from summary statistics



Math is hard. So we made it human.
Labanan ang agwat.

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Linear Regression & Correlation Worksheet — MATH 110

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6. For a bivariate data set: $r = 0.5$, $s_x = 4$, $s_y = 8$, mean of $x = 24$, mean of $y = 43$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 4, \quad s_y = 8, \quad \bar{x} = 24, \quad \bar{y} = 43$$

Answer: _____

7. For a bivariate data set: $r = 0.5$, $s_x = 4$, $s_y = 12$, mean of $x = 13$, mean of $y = 40$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 4, \quad s_y = 12, \quad \bar{x} = 13, \quad \bar{y} = 40$$

Answer: _____

8. For a bivariate data set: $r = 0.5$, $s_x = 2$, $s_y = 8$, mean of $x = 19$, mean of $y = 25$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 2, \quad s_y = 8, \quad \bar{x} = 19, \quad \bar{y} = 25$$

Answer: _____

9. For a bivariate data set: $r = 0.8$, $s_x = 2$, $s_y = 8$, mean of $x = 20$, mean of $y = 38$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.8, \quad s_x = 2, \quad s_y = 8, \quad \bar{x} = 20, \quad \bar{y} = 38$$

Answer: _____

10. For a bivariate data set: $r = 0.5$, $s_x = 2$, $s_y = 12$, mean of $x = 24$, mean of $y = 46$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 2, \quad s_y = 12, \quad \bar{x} = 24, \quad \bar{y} = 46$$

Answer: _____

Prediction from a Regression Equation

11. A regression equation is $\hat{y} = 15 + 9x$. Predict y when $x = 22$.

$$\hat{y} = 15 + 9x, \quad x = 22$$

Answer: _____

12. A regression equation is $\hat{y} = 17 + 5x$. Predict y when $x = 16$.

$$\hat{y} = 17 + 5x, \quad x = 16$$

Answer: _____



13. A regression equation is $\hat{y} = 19 + 1x$. Predict y when $x = 20$.

$$\hat{y} = 19 + 1x, \quad x = 20$$

Answer: _____

14. A regression equation is $\hat{y} = 20 + 3x$. Predict y when $x = 7$.

$$\hat{y} = 20 + 3x, \quad x = 7$$

Answer: _____

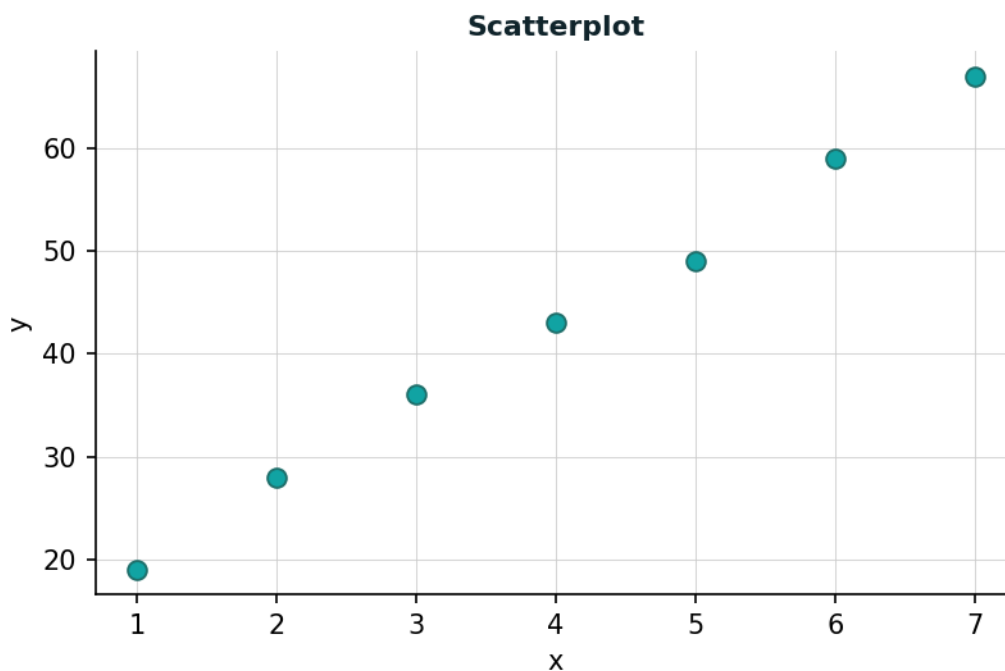
15. A regression equation is $\hat{y} = 16 + 9x$. Predict y when $x = 8$.

$$\hat{y} = 16 + 9x, \quad x = 8$$

Answer: _____

Describing a Scatterplot

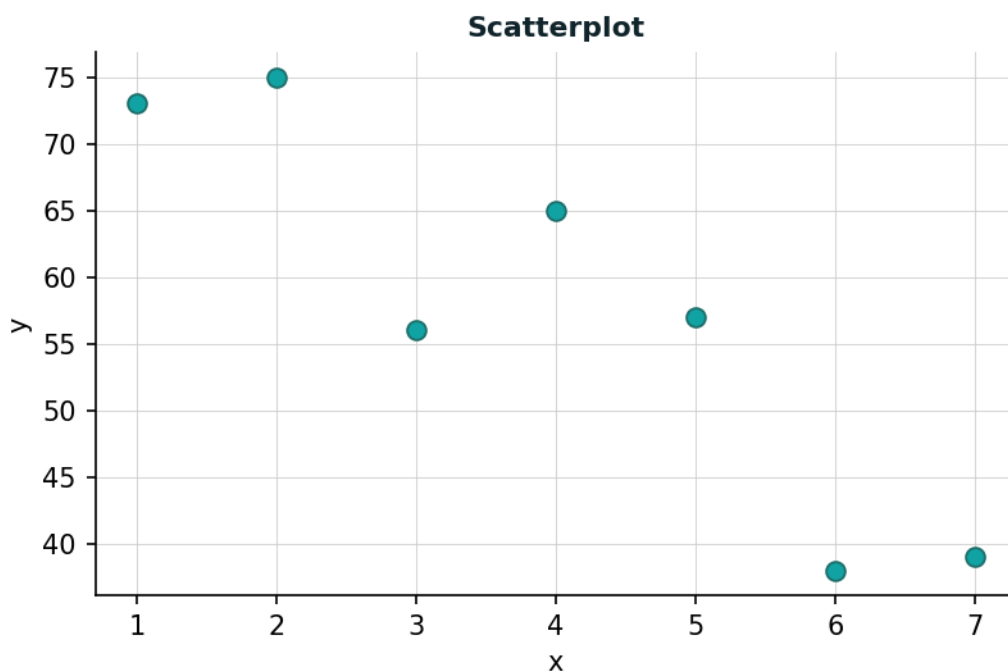
16. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

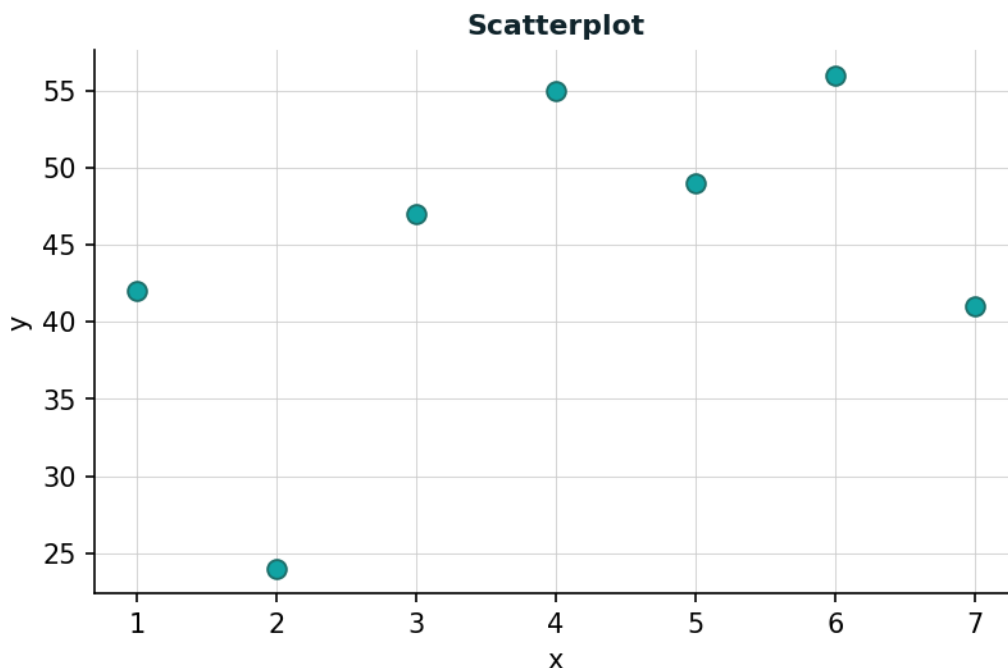


17. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

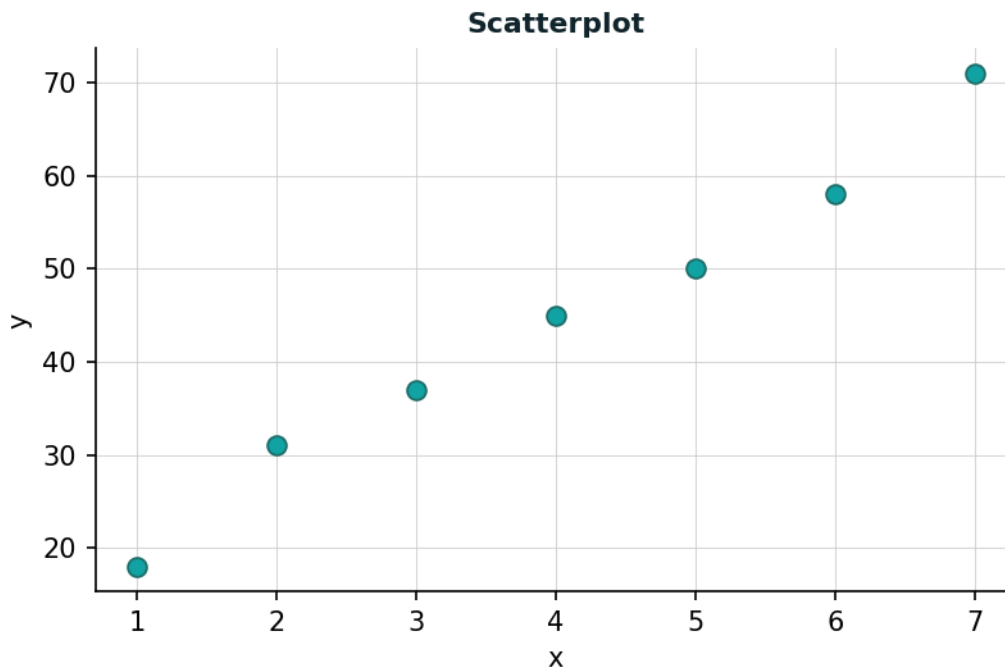
18. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

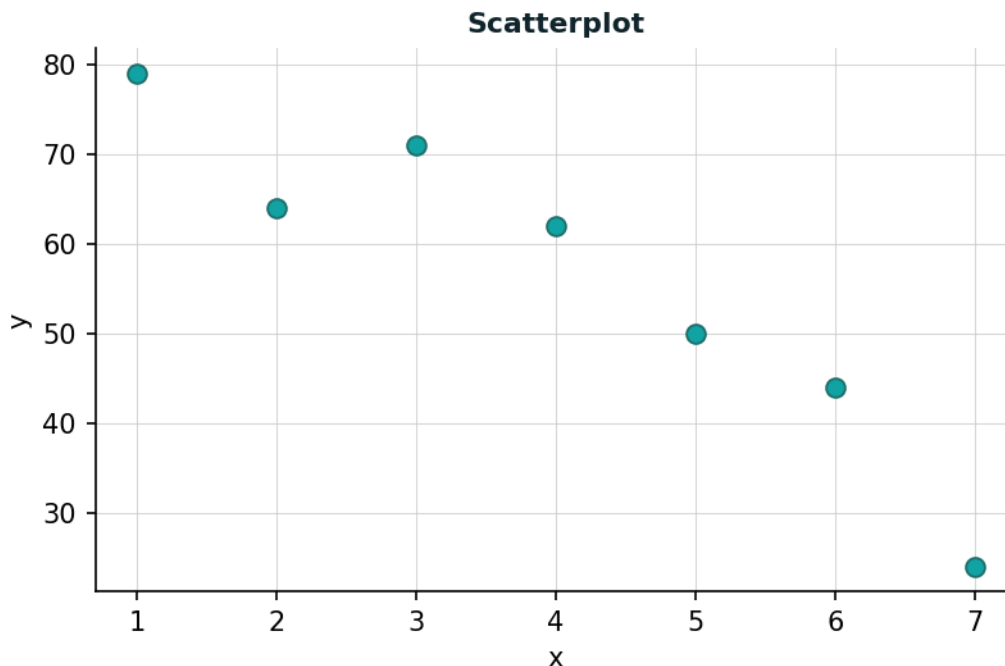


19. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

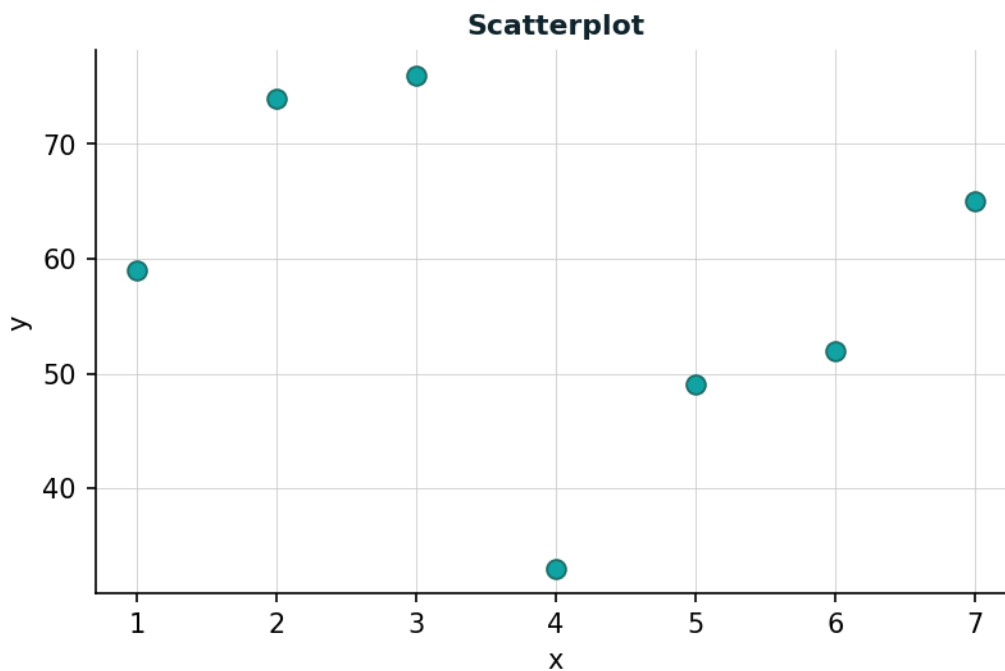
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Answer: _____

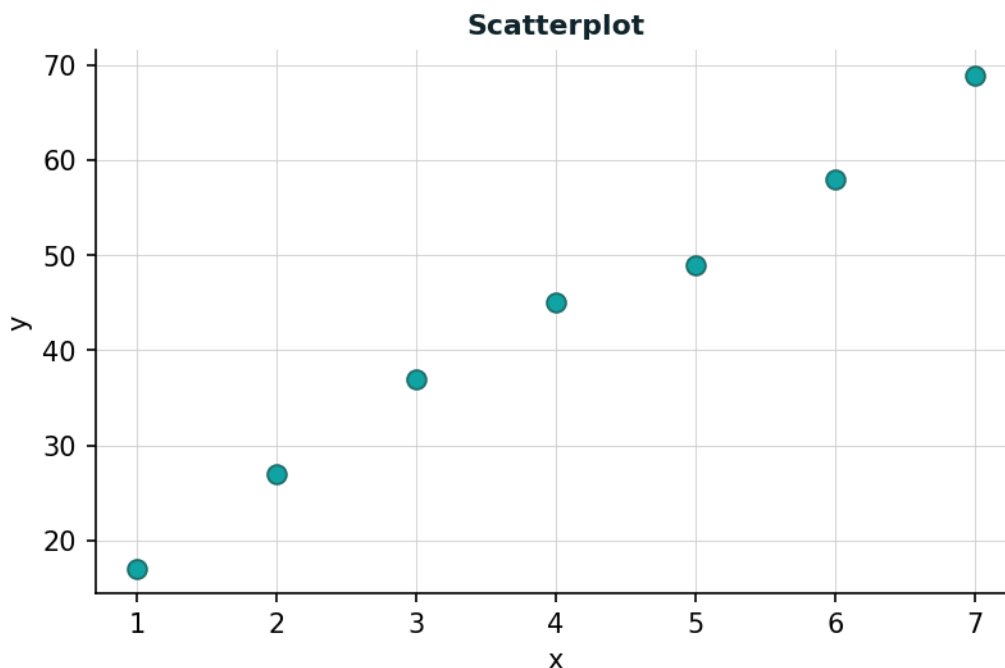


21. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

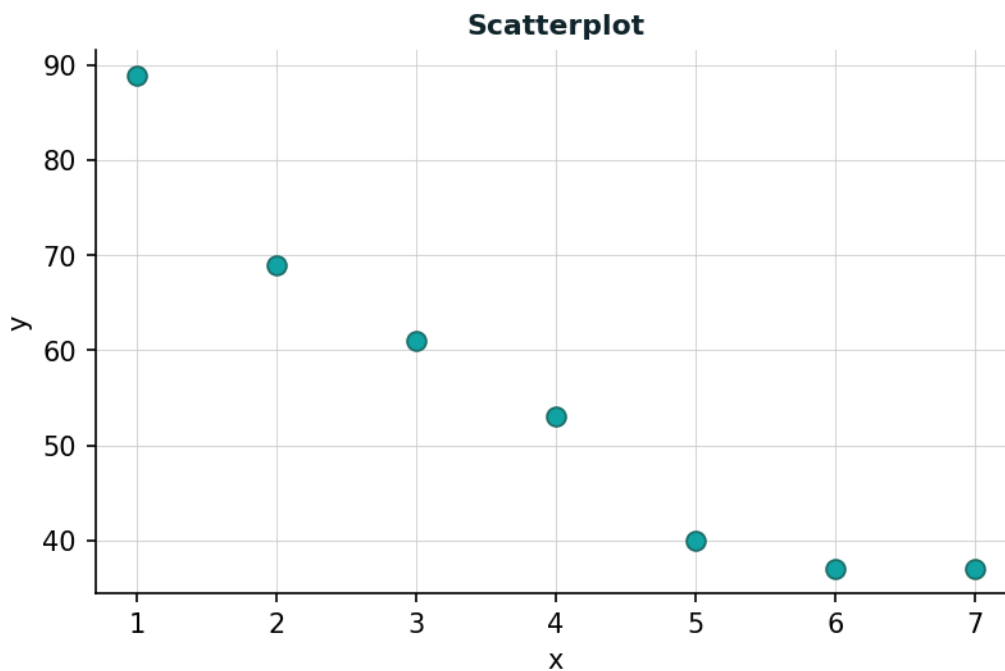
22. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

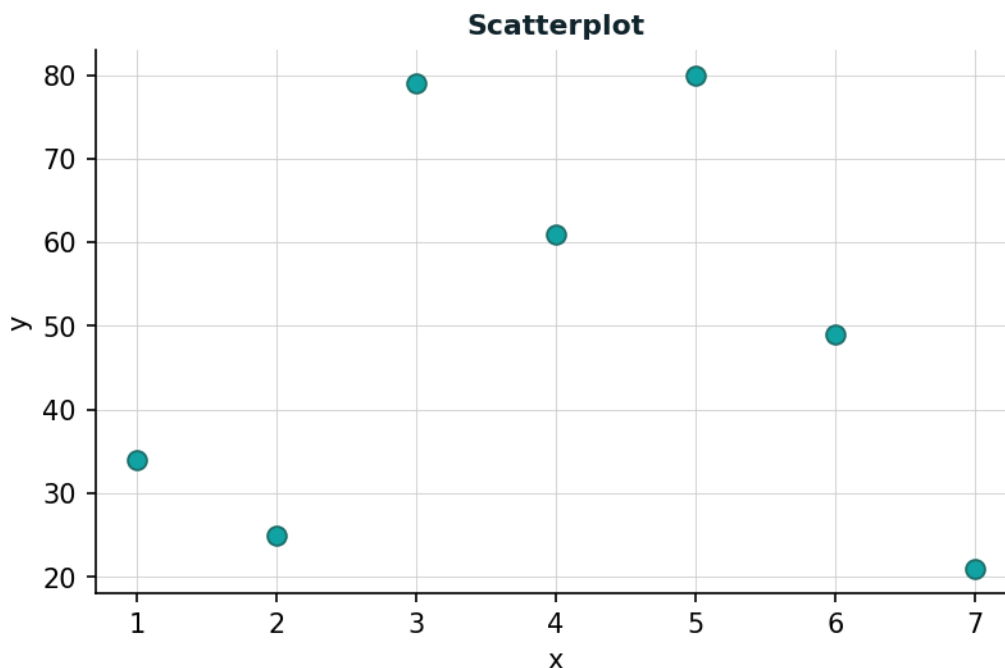


23. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

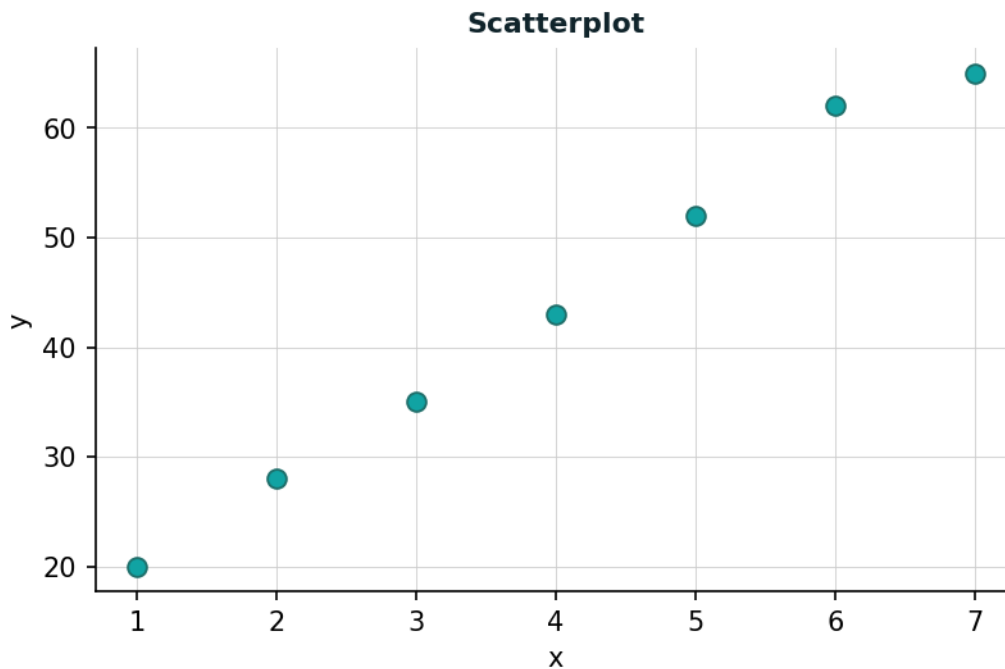
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Answer: _____

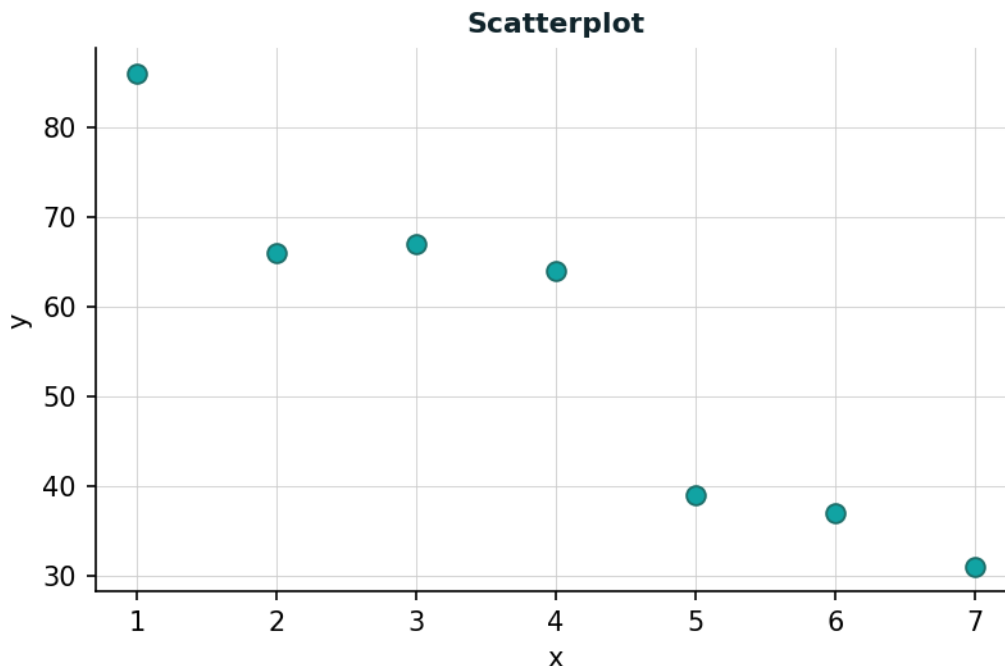


25. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

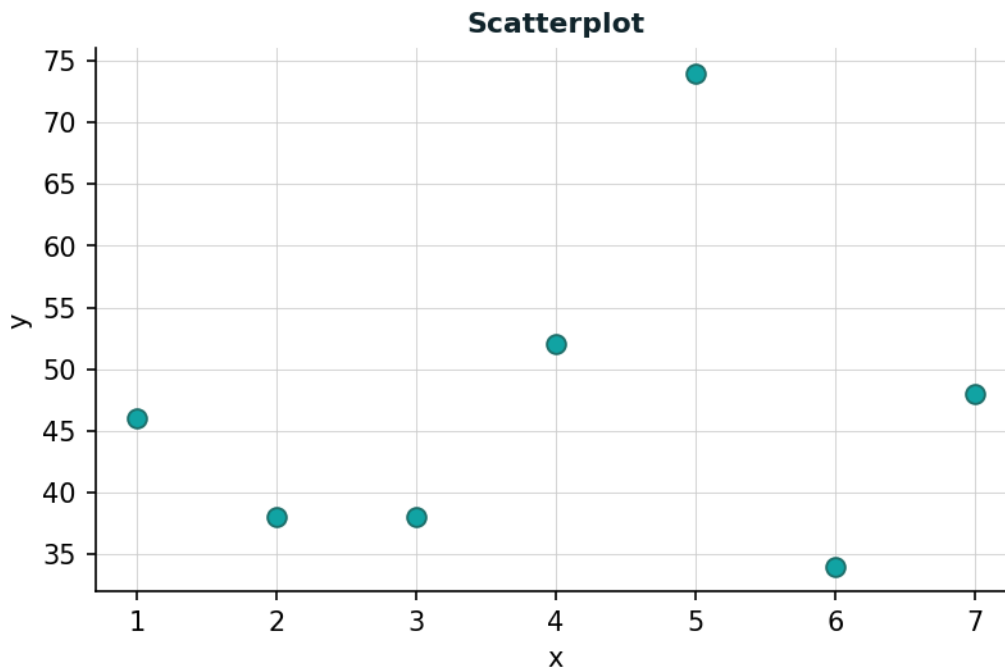
26. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

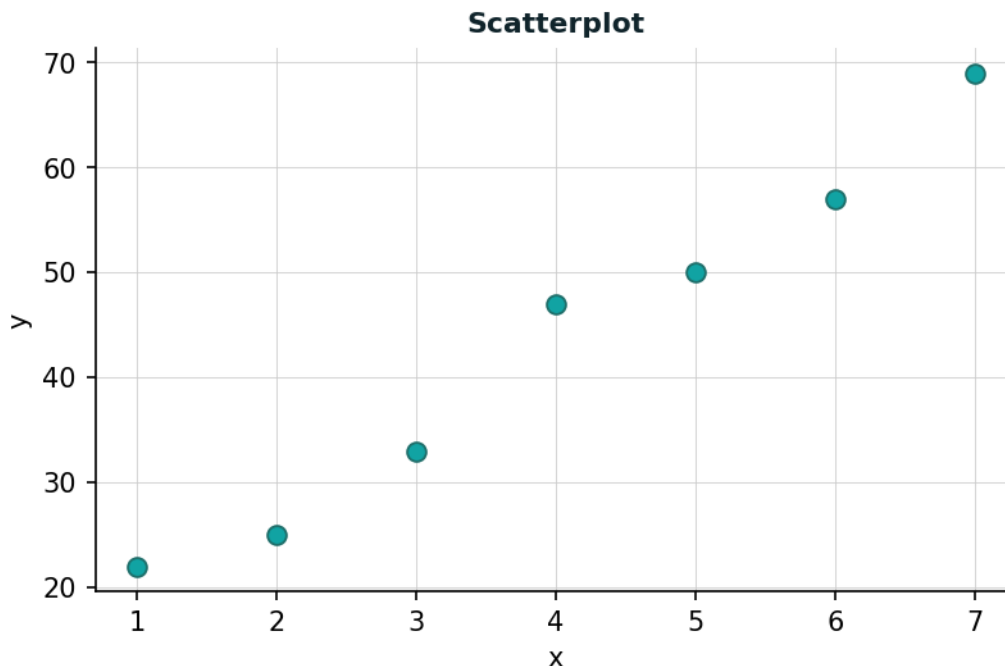


27. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

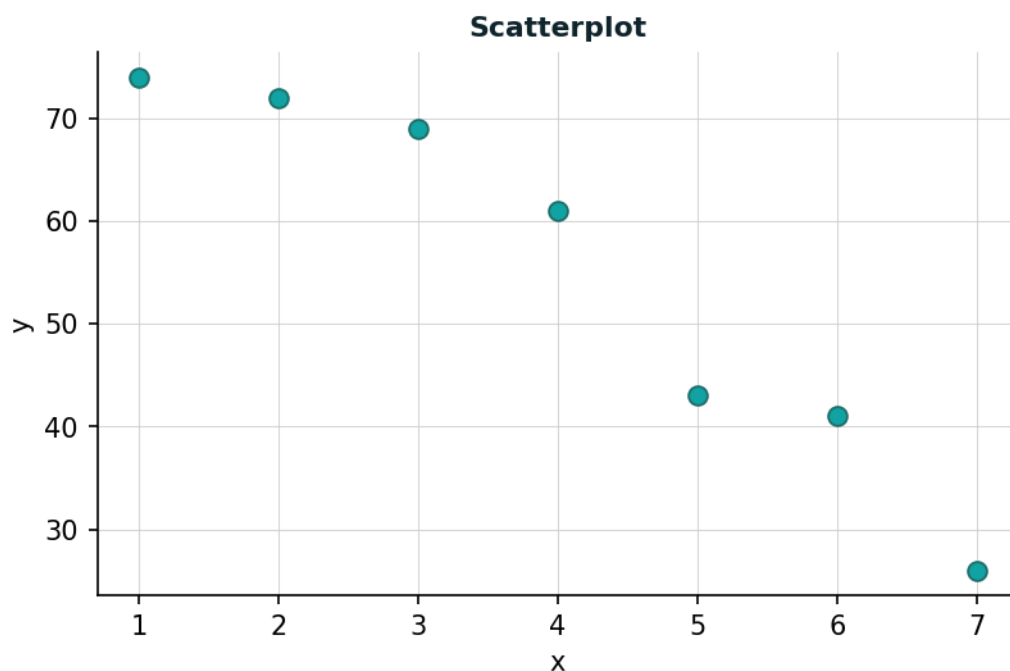
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Answer: _____

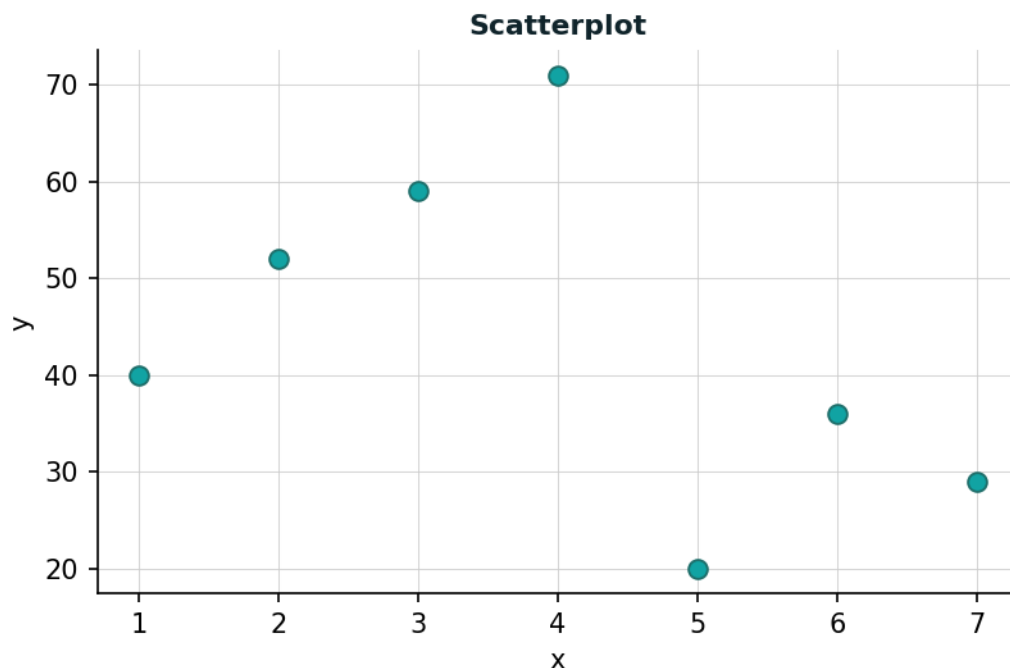


29. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____

30. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Describing a Scatterplot, Least-squares regression line from summary statistics, Correlation Coefficient Interpretation, Prediction from a Regression Equation. All answers verified by independent computation.

Solutions



Correlation Coefficient Interpretation

1. A study yields a correlation coefficient of $r = 0.78$. Describe the direction and strength of the linear relationship.

$$r = 0.78$$

- Sign of r : $r=0.78$ is positive (positive r means positive association, negative means negative).
- $|r| = 0.78$. Strength: $|r| < 0.3$ is weak, 0.3 to 0.7 is moderate, > 0.7 is strong.
- This is a strong positive linear relationship.

Answer: $r = 0.78$: positive, strong relationship

2. A study yields a correlation coefficient of $r = 0.54$. Describe the direction and strength of the linear relationship.

$$r = 0.54$$

- Sign of r : $r=0.54$ is positive (positive r means positive association, negative means negative).
- $|r| = 0.54$. Strength: $|r| < 0.3$ is weak, 0.3 to 0.7 is moderate, > 0.7 is strong.
- This is a moderate positive linear relationship.

Answer: $r = 0.54$: positive, moderate relationship

3. A study yields a correlation coefficient of $r = -0.65$. Describe the direction and strength of the linear relationship.

$$r = -0.65$$

- Sign of r : $r=-0.65$ is negative (positive r means positive association, negative means negative).
- $|r| = 0.65$. Strength: $|r| < 0.3$ is weak, 0.3 to 0.7 is moderate, > 0.7 is strong.
- This is a moderate negative linear relationship.

Answer: $r = -0.65$: negative, moderate relationship

4. A study yields a correlation coefficient of $r = -0.28$. Describe the direction and strength of the linear relationship.

$$r = -0.28$$

- Sign of r : $r=-0.28$ is negative (positive r means positive association, negative means negative).
- $|r| = 0.28$. Strength: $|r| < 0.3$ is weak, 0.3 to 0.7 is moderate, > 0.7 is strong.
- This is a weak negative linear relationship.

Answer: $r = -0.28$: negative, weak relationship

5. A study yields a correlation coefficient of $r = -0.92$. Describe the direction and strength of the linear relationship.

$$r = -0.92$$

- Sign of r : $r=-0.92$ is negative (positive r means positive association, negative means negative).
- $|r| = 0.92$. Strength: $|r| < 0.3$ is weak, 0.3 to 0.7 is moderate, > 0.7 is strong.
- This is a strong negative linear relationship.

Answer: $r = -0.92$: negative, strong relationship



Least-squares regression line from summary statistics

6. For a bivariate data set: $r = 0.5$, $s_x = 4$, $s_y = 8$, mean of $x = 24$, mean of $y = 43$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 4, \quad s_y = 8, \quad \bar{x} = 24, \quad \bar{y} = 43$$

$$\rightarrow \text{Slope } b = r \times (s_y/s_x) = 0.5 \times (8/4) = 1.0.$$

$$\rightarrow \text{Intercept } a = (\text{mean } y) - b \times (\text{mean } x) = 43 - (1.0)(24) = 19.0.$$

$$\rightarrow \text{Regression line: } \hat{y} = 19.0 + 1.0x \text{ (r-squared} = 0.25).$$

Answer: $b = r \frac{s_y}{s_x} = 1.0, \quad a = \bar{y} - b\bar{x} = 19.0 \Rightarrow \hat{y} = 19.0 + 1.0x$

7. For a bivariate data set: $r = 0.5$, $s_x = 4$, $s_y = 12$, mean of $x = 13$, mean of $y = 40$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 4, \quad s_y = 12, \quad \bar{x} = 13, \quad \bar{y} = 40$$

$$\rightarrow \text{Slope } b = r \times (s_y/s_x) = 0.5 \times (12/4) = 1.5.$$

$$\rightarrow \text{Intercept } a = (\text{mean } y) - b \times (\text{mean } x) = 40 - (1.5)(13) = 20.5.$$

$$\rightarrow \text{Regression line: } \hat{y} = 20.5 + 1.5x \text{ (r-squared} = 0.25).$$

Answer: $b = r \frac{s_y}{s_x} = 1.5, \quad a = \bar{y} - b\bar{x} = 20.5 \Rightarrow \hat{y} = 20.5 + 1.5x$

8. For a bivariate data set: $r = 0.5$, $s_x = 2$, $s_y = 8$, mean of $x = 19$, mean of $y = 25$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 2, \quad s_y = 8, \quad \bar{x} = 19, \quad \bar{y} = 25$$

$$\rightarrow \text{Slope } b = r \times (s_y/s_x) = 0.5 \times (8/2) = 2.0.$$

$$\rightarrow \text{Intercept } a = (\text{mean } y) - b \times (\text{mean } x) = 25 - (2.0)(19) = -13.0.$$

$$\rightarrow \text{Regression line: } \hat{y} = -13.0 + 2.0x \text{ (r-squared} = 0.25).$$

Answer: $b = r \frac{s_y}{s_x} = 2.0, \quad a = \bar{y} - b\bar{x} = -13.0 \Rightarrow \hat{y} = -13.0 + 2.0x$

9. For a bivariate data set: $r = 0.8$, $s_x = 2$, $s_y = 8$, mean of $x = 20$, mean of $y = 38$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.8, \quad s_x = 2, \quad s_y = 8, \quad \bar{x} = 20, \quad \bar{y} = 38$$

$$\rightarrow \text{Slope } b = r \times (s_y/s_x) = 0.8 \times (8/2) = 3.2.$$

$$\rightarrow \text{Intercept } a = (\text{mean } y) - b \times (\text{mean } x) = 38 - (3.2)(20) = -26.0.$$

$$\rightarrow \text{Regression line: } \hat{y} = -26.0 + 3.2x \text{ (r-squared} = 0.64).$$

Answer: $b = r \frac{s_y}{s_x} = 3.2, \quad a = \bar{y} - b\bar{x} = -26.0 \Rightarrow \hat{y} = -26.0 + 3.2x$



10. For a bivariate data set: $r = 0.5$, $s_x = 2$, $s_y = 12$, mean of $x = 24$, mean of $y = 46$. Find the least-squares regression line $\hat{y} = a + bx$.

$$r = 0.5, \quad s_x = 2, \quad s_y = 12, \quad \bar{x} = 24, \quad \bar{y} = 46$$

$$\rightarrow \text{Slope } b = r \times (s_y/s_x) = 0.5 \times (12/2) = 3.0.$$

$$\rightarrow \text{Intercept } a = (\text{mean } y) - b \times (\text{mean } x) = 46 - (3.0)(24) = -26.0.$$

$$\rightarrow \text{Regression line: } \hat{y} = -26.0 + 3.0x \text{ (r-squared} = 0.25\text{)}.$$

Answer: $b = r \frac{s_y}{s_x} = 3.0, \quad a = \bar{y} - b\bar{x} = -26.0 \Rightarrow \hat{y} = -26.0 + 3.0x$



Prediction from a Regression Equation

11. A regression equation is $\hat{y} = 15 + 9x$. Predict y when $x = 22$.

$$\hat{y} = 15 + 9x, \quad x = 22$$

→ Substitute $x = 22$ into $\hat{y} = 15 + 9x$.

$$\rightarrow \hat{y} = 15 + 9(22) = 213.$$

Answer: $\hat{y} = 15 + 9(22) = 213$

12. A regression equation is $\hat{y} = 17 + 5x$. Predict y when $x = 16$.

$$\hat{y} = 17 + 5x, \quad x = 16$$

→ Substitute $x = 16$ into $\hat{y} = 17 + 5x$.

$$\rightarrow \hat{y} = 17 + 5(16) = 97.$$

Answer: $\hat{y} = 17 + 5(16) = 97$

13. A regression equation is $\hat{y} = 19 + 1x$. Predict y when $x = 20$.

$$\hat{y} = 19 + 1x, \quad x = 20$$

→ Substitute $x = 20$ into $\hat{y} = 19 + 1x$.

$$\rightarrow \hat{y} = 19 + 1(20) = 39.$$

Answer: $\hat{y} = 19 + 1(20) = 39$

14. A regression equation is $\hat{y} = 20 + 3x$. Predict y when $x = 7$.

$$\hat{y} = 20 + 3x, \quad x = 7$$

→ Substitute $x = 7$ into $\hat{y} = 20 + 3x$.

$$\rightarrow \hat{y} = 20 + 3(7) = 41.$$

Answer: $\hat{y} = 20 + 3(7) = 41$

15. A regression equation is $\hat{y} = 16 + 9x$. Predict y when $x = 8$.

$$\hat{y} = 16 + 9x, \quad x = 8$$

→ Substitute $x = 8$ into $\hat{y} = 16 + 9x$.

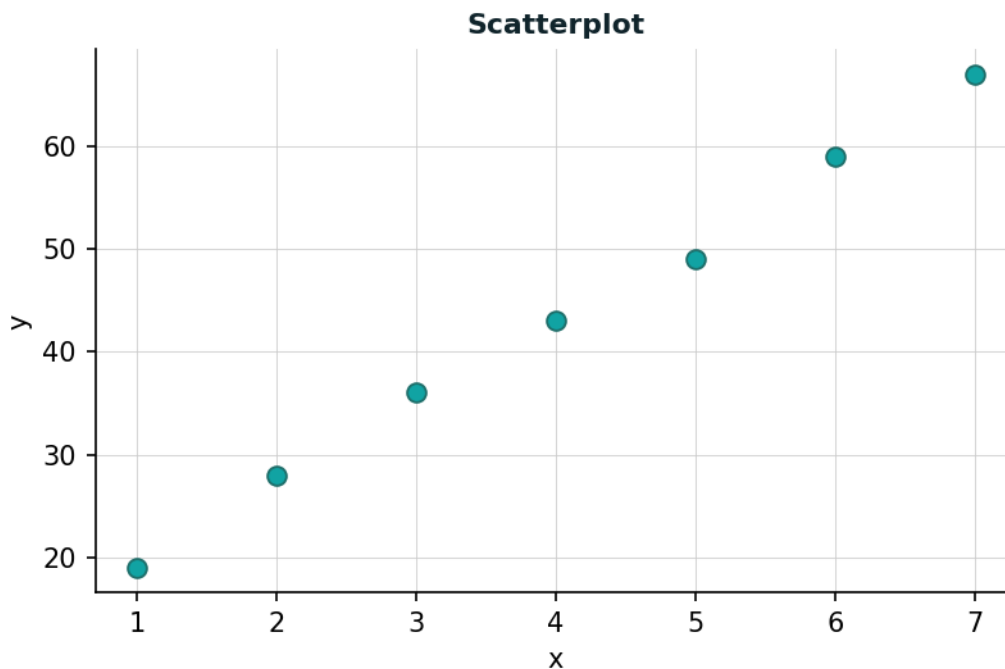
$$\rightarrow \hat{y} = 16 + 9(8) = 88.$$

Answer: $\hat{y} = 16 + 9(8) = 88$



Describing a Scatterplot

16. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points rise left to right -> positive.

→ Form: points cluster around a line -> linear.

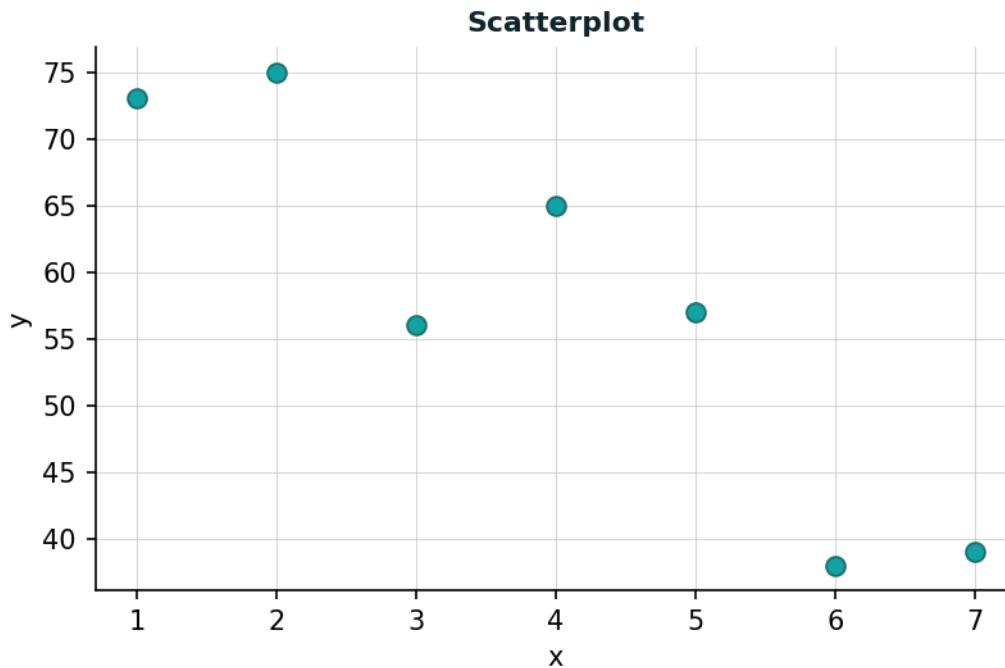
→ Strength: tight clustering -> strong.

→ Answer: A) positive, linear, strong.

Answer: A)positive, linear, strong



17. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points decline left to right -> negative.

→ Form: points loosely follow a line -> linear.

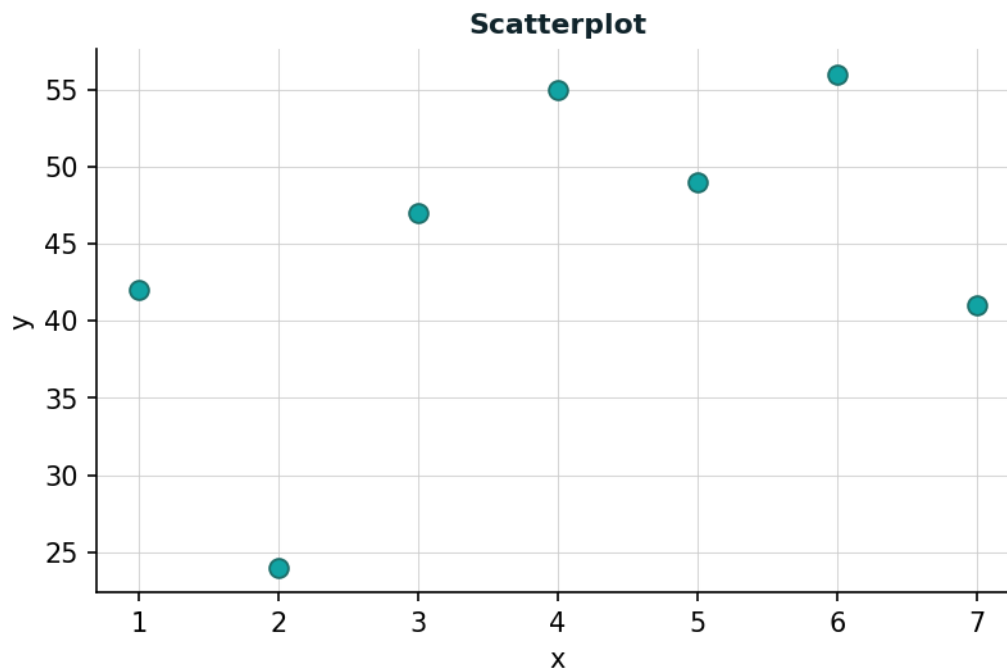
→ Strength: moderate scatter -> moderate.

→ Answer: B) negative, linear, moderate.

Answer: B)negative, linear, moderate



18. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



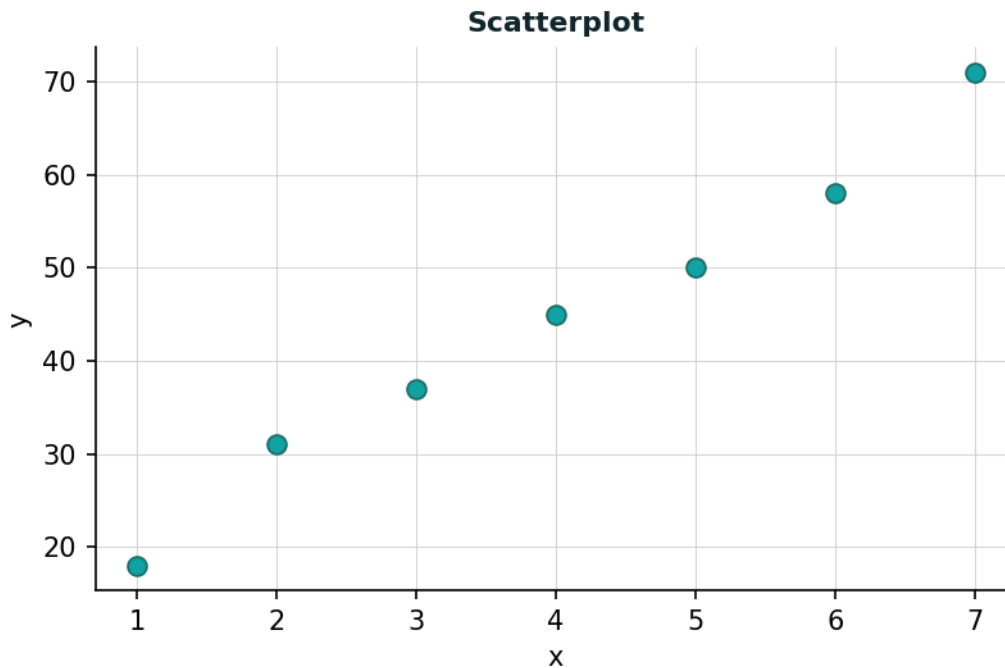
→ No pattern is visible in the scatterplot -> no association.

→ Answer: C) no association.

Answer: C)noassociation



19. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points rise left to right -> positive.

→ Form: points cluster around a line -> linear.

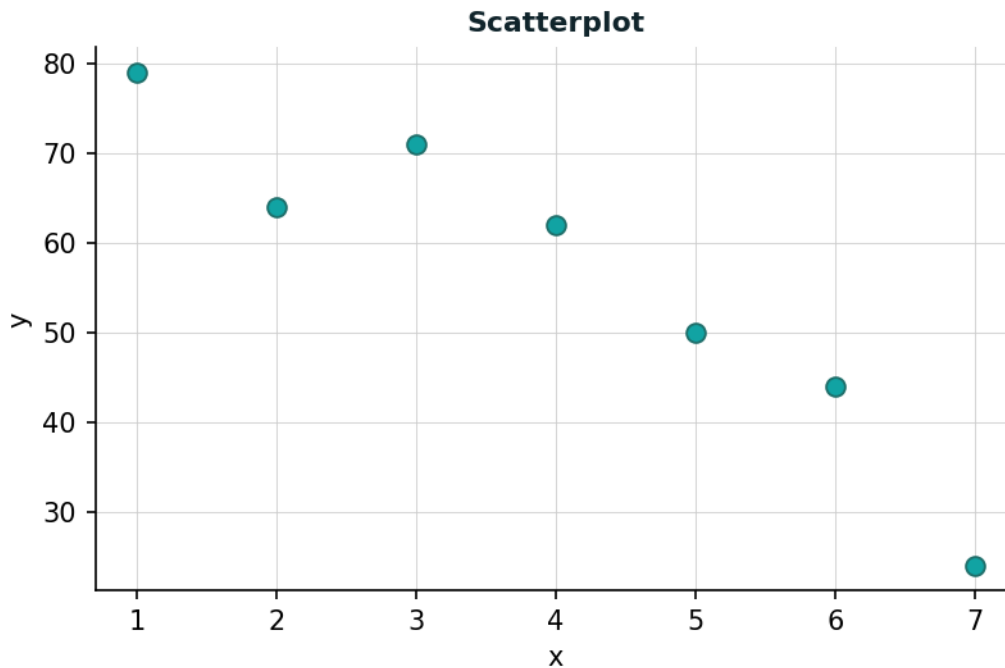
→ Strength: tight clustering -> strong.

→ Answer: A) positive, linear, strong.

Answer: A)positive, linear, strong



20. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points decline left to right -> negative.

→ Form: points loosely follow a line -> linear.

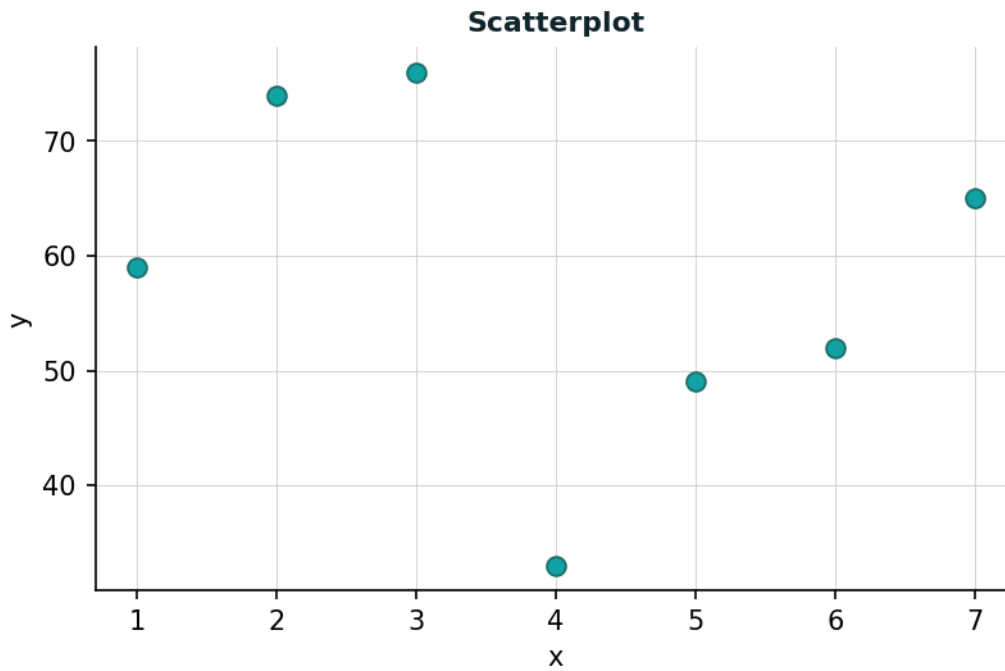
→ Strength: moderate scatter -> moderate.

→ Answer: B) negative, linear, moderate.

Answer: B)negative, linear, moderate



21. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



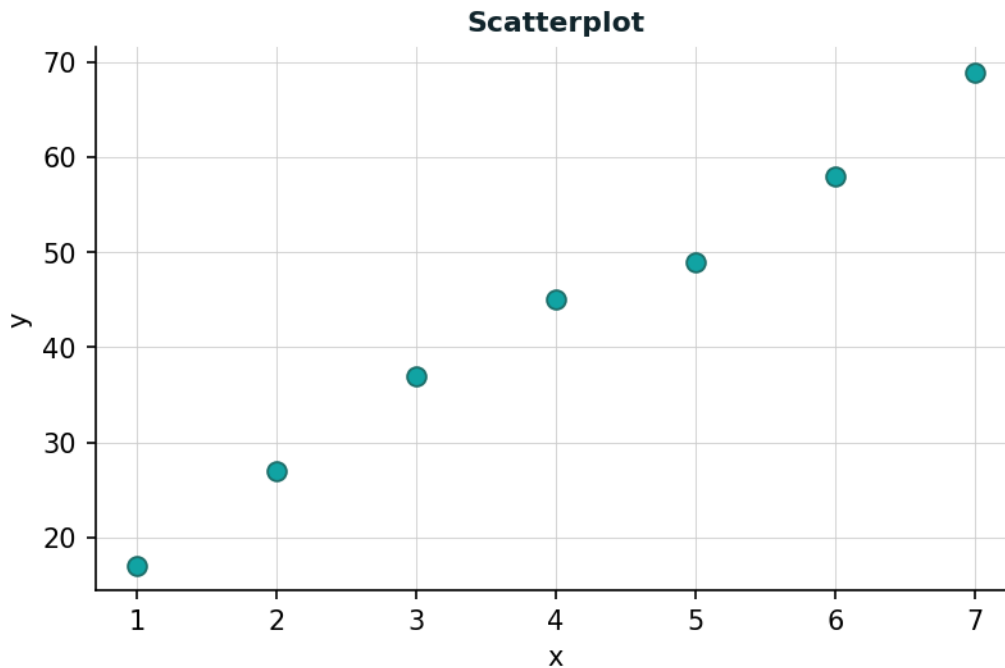
→ No pattern is visible in the scatterplot -> no association.

→ Answer: C) no association.

Answer: C)noassociation



22. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points rise left to right -> positive.

→ Form: points cluster around a line -> linear.

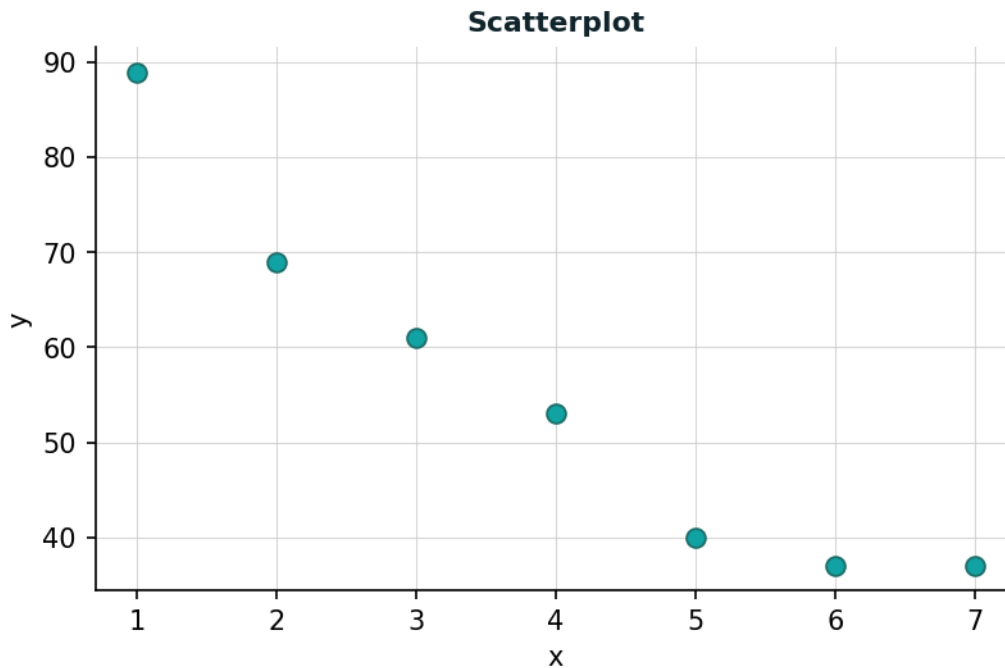
→ Strength: tight clustering -> strong.

→ Answer: A) positive, linear, strong.

Answer: A)positive, linear, strong



23. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points decline left to right -> negative.

→ Form: points loosely follow a line -> linear.

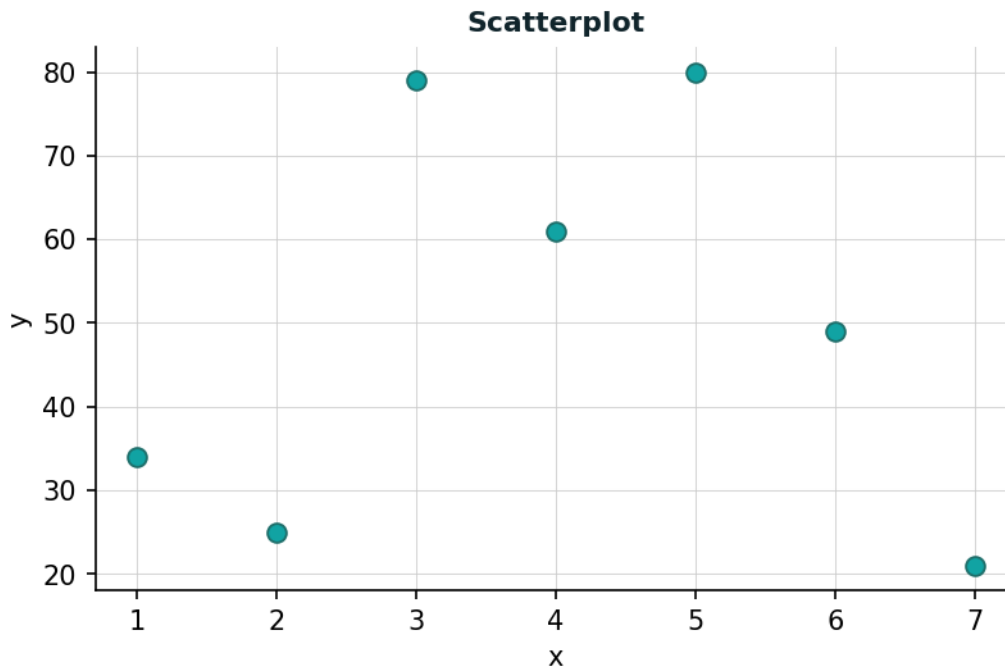
→ Strength: moderate scatter -> moderate.

→ Answer: B) negative, linear, moderate.

Answer: B)negative, linear, moderate



24. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



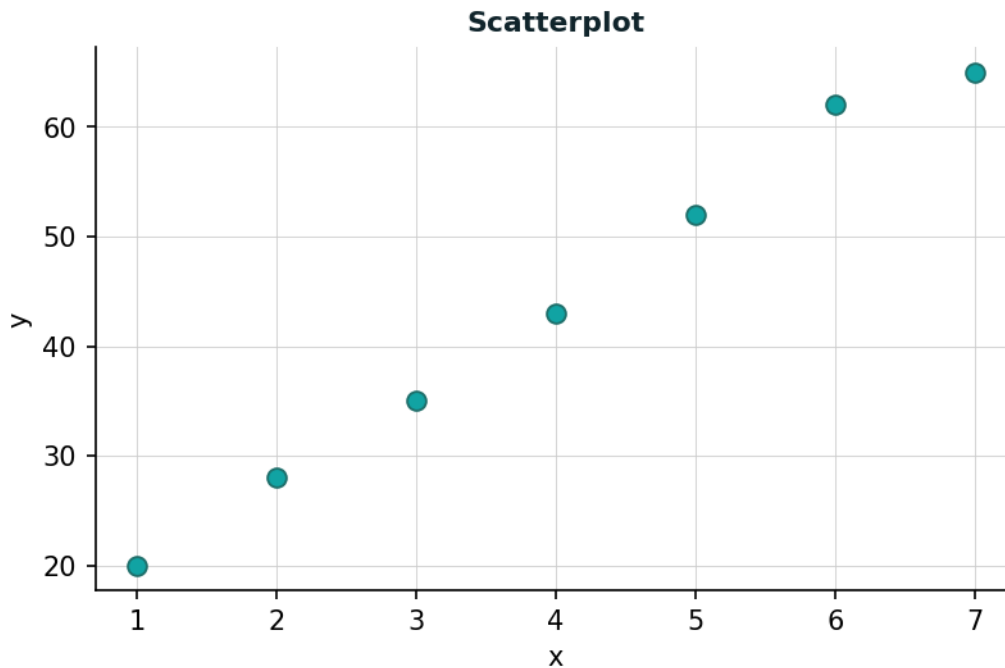
→ No pattern is visible in the scatterplot -> no association.

→ Answer: C) no association.

Answer: C)noassociation



25. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points rise left to right -> positive.

→ Form: points cluster around a line -> linear.

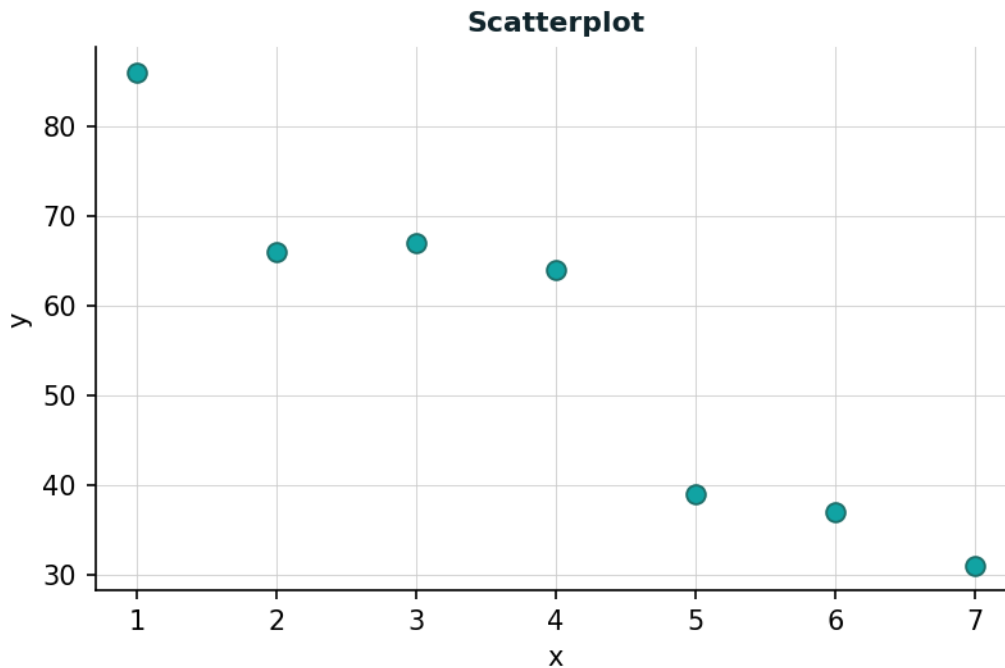
→ Strength: tight clustering -> strong.

→ Answer: A) positive, linear, strong.

Answer: A)positive, linear, strong



26. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points decline left to right -> negative.

→ Form: points loosely follow a line -> linear.

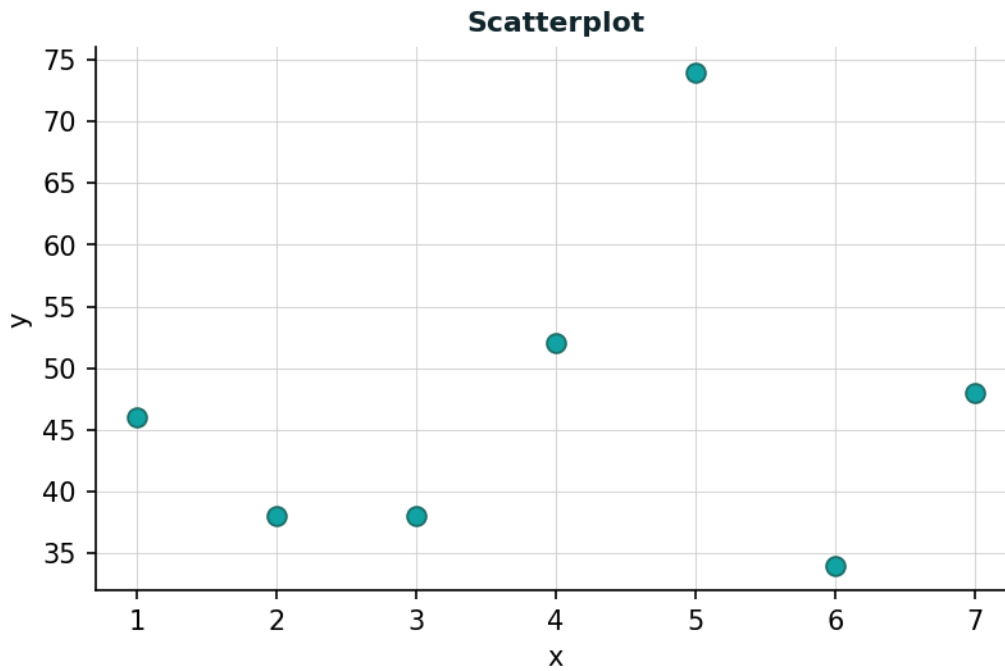
→ Strength: moderate scatter -> moderate.

→ Answer: B) negative, linear, moderate.

Answer: B)negative, linear, moderate



27. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



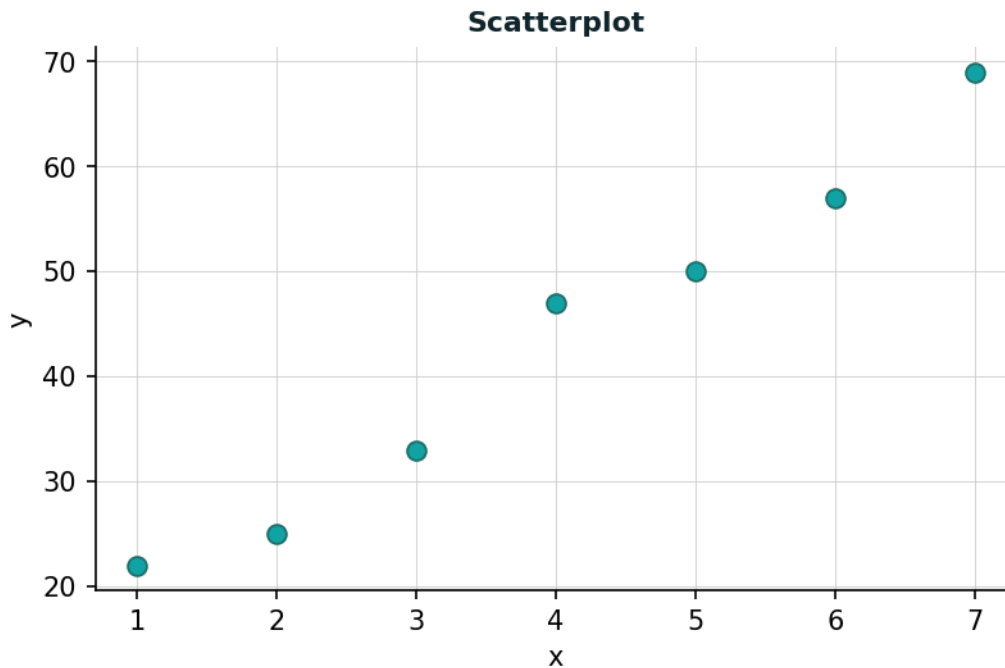
→ No pattern is visible in the scatterplot -> no association.

→ Answer: C) no association.

Answer: C)noassociation



28. A scatterplot shows: points clustered tightly around a line rising from left to right, with no unusual values. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points rise left to right -> positive.

→ Form: points cluster around a line -> linear.

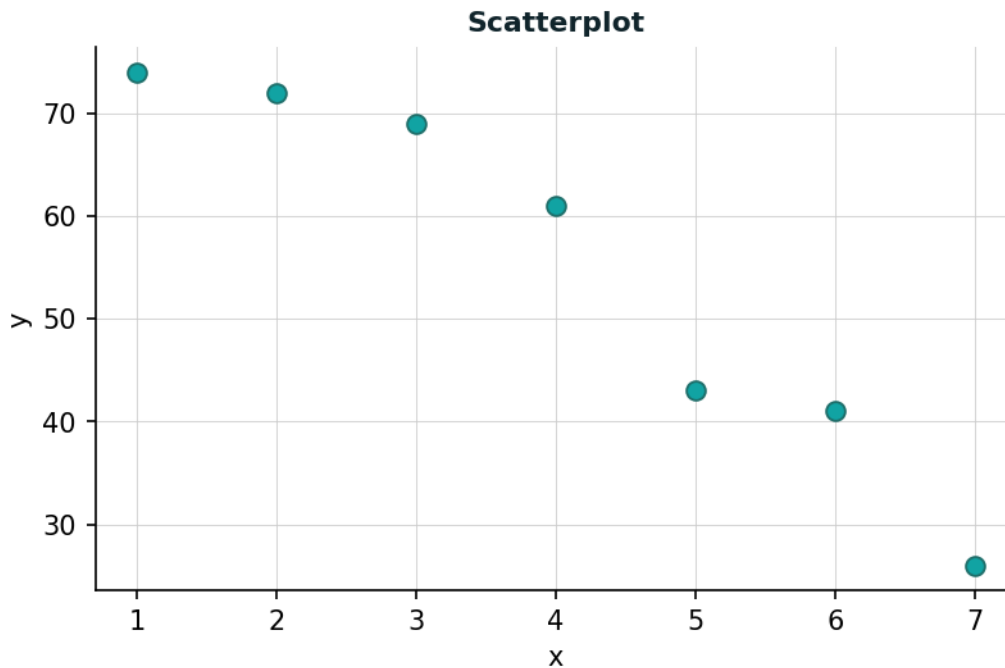
→ Strength: tight clustering -> strong.

→ Answer: A) positive, linear, strong.

Answer: A)positive, linear, strong



29. A scatterplot shows: points that fall loosely along a line declining from left to right, with moderate scatter. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ Direction: points decline left to right -> negative.

→ Form: points loosely follow a line -> linear.

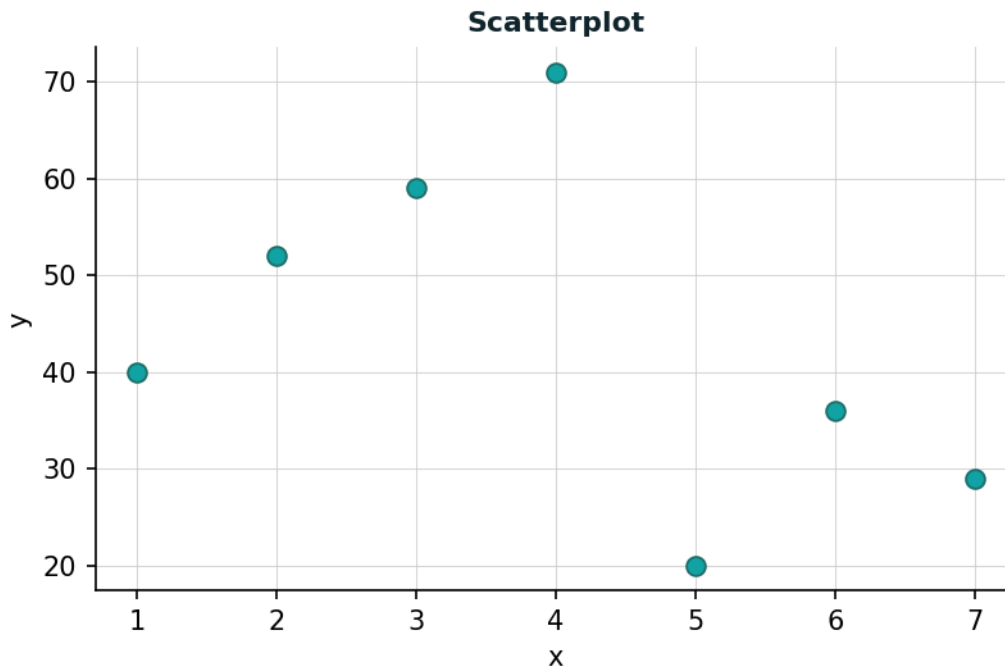
→ Strength: moderate scatter -> moderate.

→ Answer: B) negative, linear, moderate.

Answer: B) negative, linear, moderate



30. A scatterplot shows: points scattered randomly with no discernible pattern or trend. What are the direction, form, and strength of the association? Options: A) positive, linear, strong B) negative, linear, moderate C) no association



→ No pattern is visible in the scatterplot -> no association.

→ Answer: C) no association.

Answer: C)noassociation

